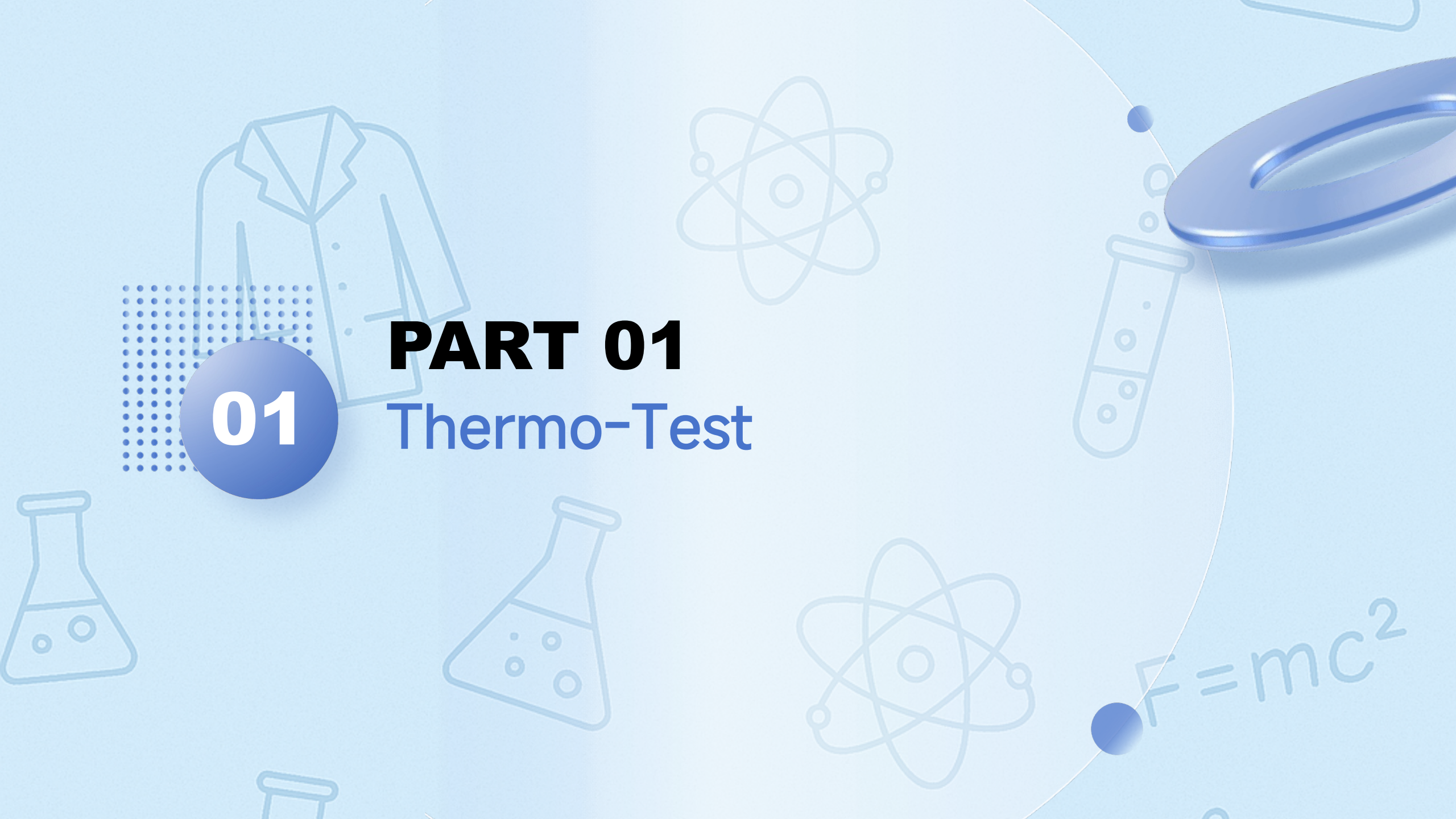




Lesson Challenges

$$E=mc^2$$



01

PART 01

Thermo-Test

$$E=mc^2$$

Survival Challenge

Imagine you drop a desert lizard and an arctic fox into a snowstorm.



- Which one is more likely to survive and why?
- What internal systems help one animal survive better than the other?

Warm Blood Wins the Cold

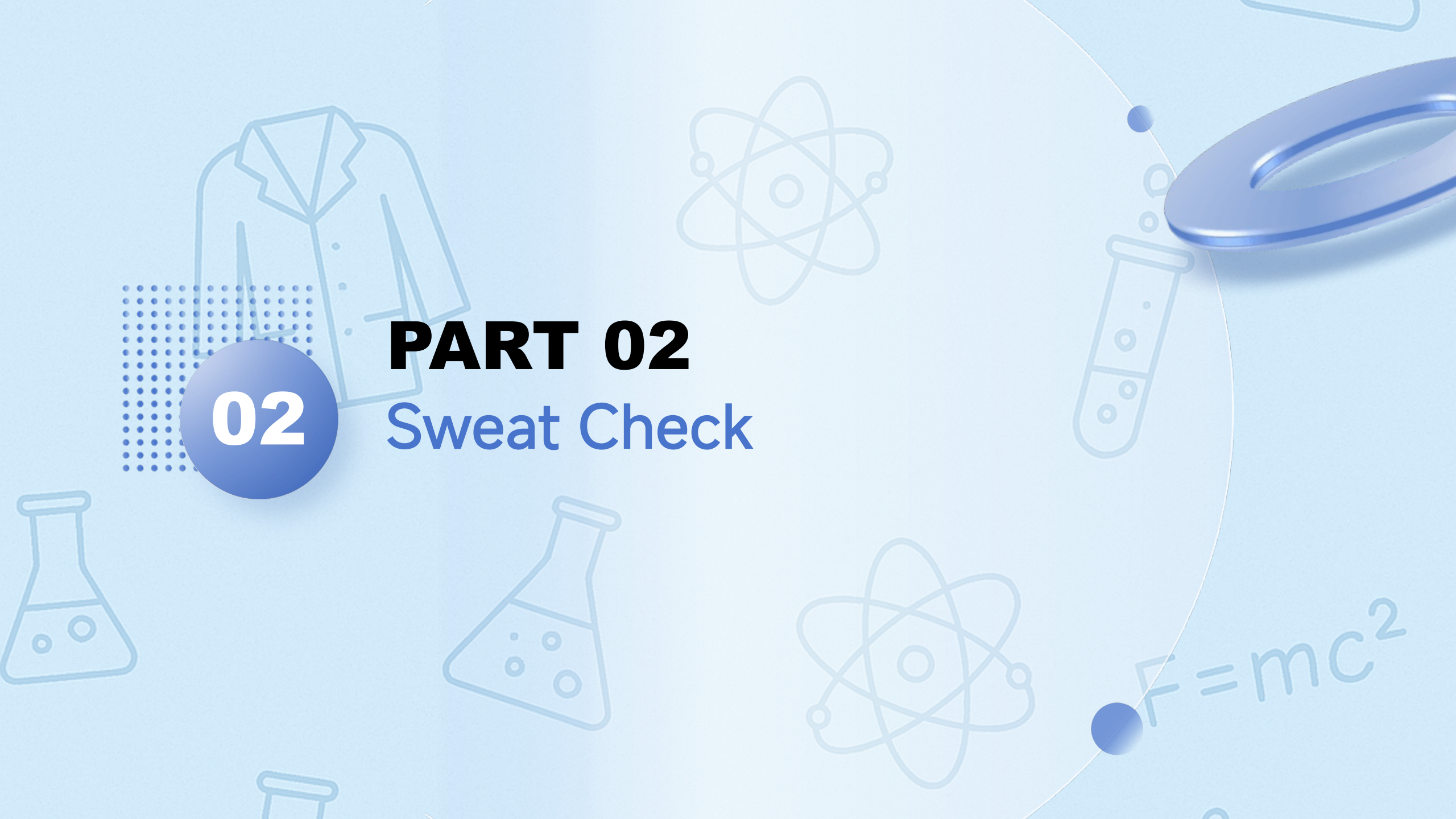


Endothermic Advantage

The arctic fox's endothermic nature allows it to generate heat internally, maintaining a constant body temperature even in freezing conditions. This metabolic heat production is essential for survival in the harsh Arctic environment.

Ectothermic Limitation

The desert lizard, being ectothermic, relies on external heat sources to regulate its body temperature. In a snowstorm, it cannot generate enough heat to stay active, leading to immobility and potential death.



02

PART 02

Sweat Check

Sweaty Alex vs Dry Max

Sweating Mechanism

Alex is sweating after playing soccer on a hot day. Max is not sweating at all.

- What's happening inside both their bodies?
- Which one is at risk of overheating?
- What role does the hypothalamus play?



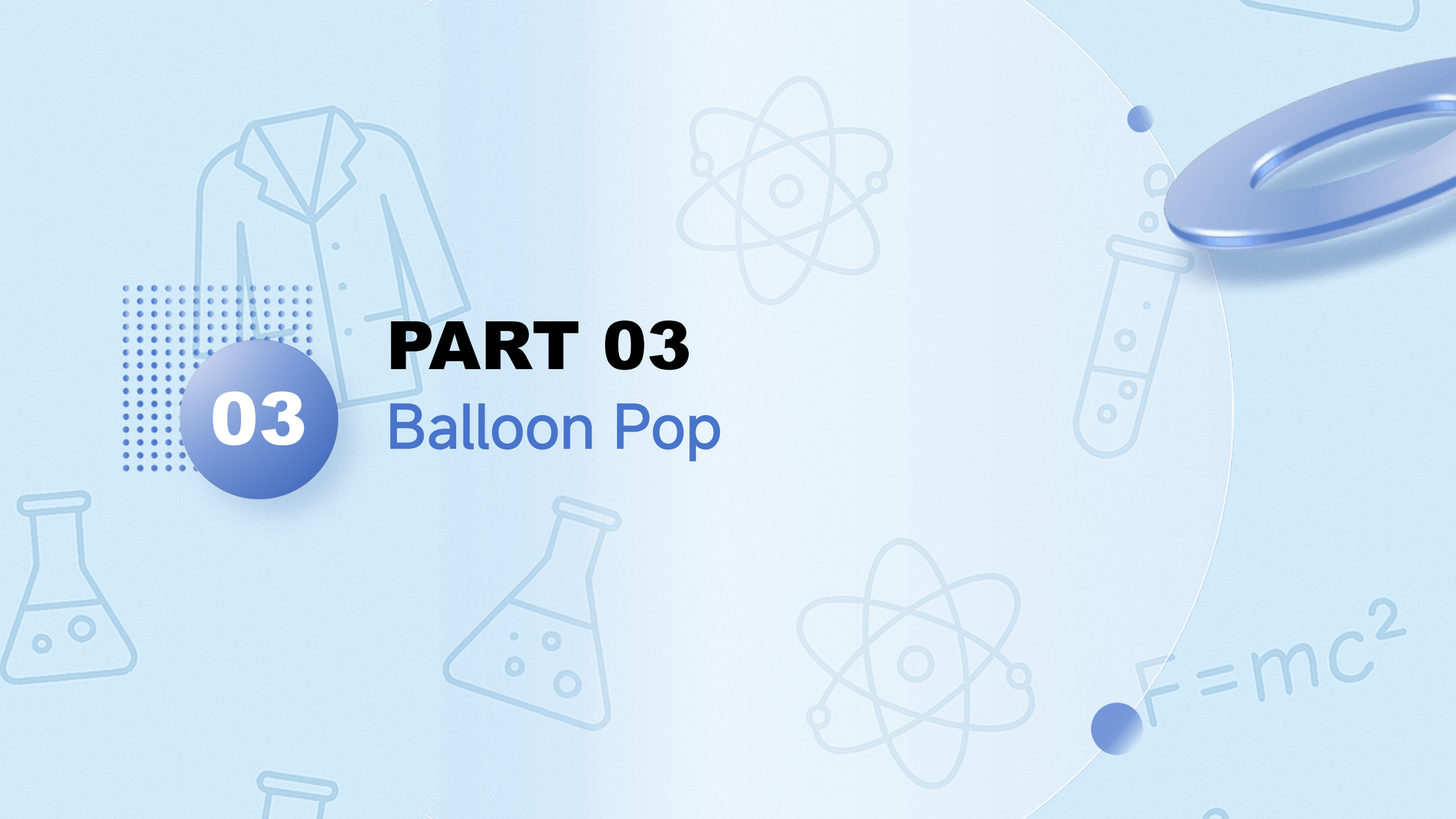
Sweating Is Life-Saving



Sweating and Survival

Sweating, triggered by the hypothalamus, helps Alex cool down. Max, not sweating, may be in danger of heat exhaustion or heat stroke.

Sweating is a thermoregulatory process, controlled by the hypothalamus, and it is crucial for maintaining homeostasis and ensuring survival in hot conditions.



03

PART 03

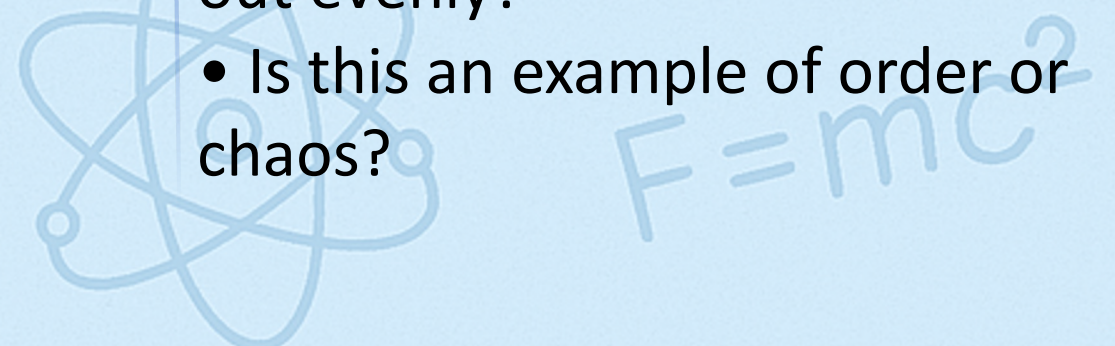
Balloon Pop



Perfume Balloon Burst

You pop a balloon filled with perfume in the middle of a room.

- Why does the smell spread out evenly?
- Is this an example of order or chaos?



Entropy in Action

Molecular Movement

Perfume molecules move randomly, spreading from the source to fill the room in the process of diffusion. This random motion ensures that the scent is evenly distributed, demonstrating the natural tendency of systems to move toward equilibrium.

Concentration Gradient

Diffusion occurs along a concentration gradient, where molecules move from high to low concentration. This process continues until the concentration is uniform throughout the space.





04

PART 06

High Altitude



High altitude breathing

Why do people hyperventilate (breathe both faster and deeper than normal) when they climb to a top of a mountain?

$$E=mc^2$$

Altitude Oxygen Crisis



Immediate Response

At high altitude there's less oxygen, so the body responds with hyperventilation to increase oxygen intake. This immediate reaction helps counteract the lower oxygen levels in the environment.



Long-Term Adaptation

Over time, the body increases red blood cell production. This long-term adaptation improves oxygen delivery to tissues.





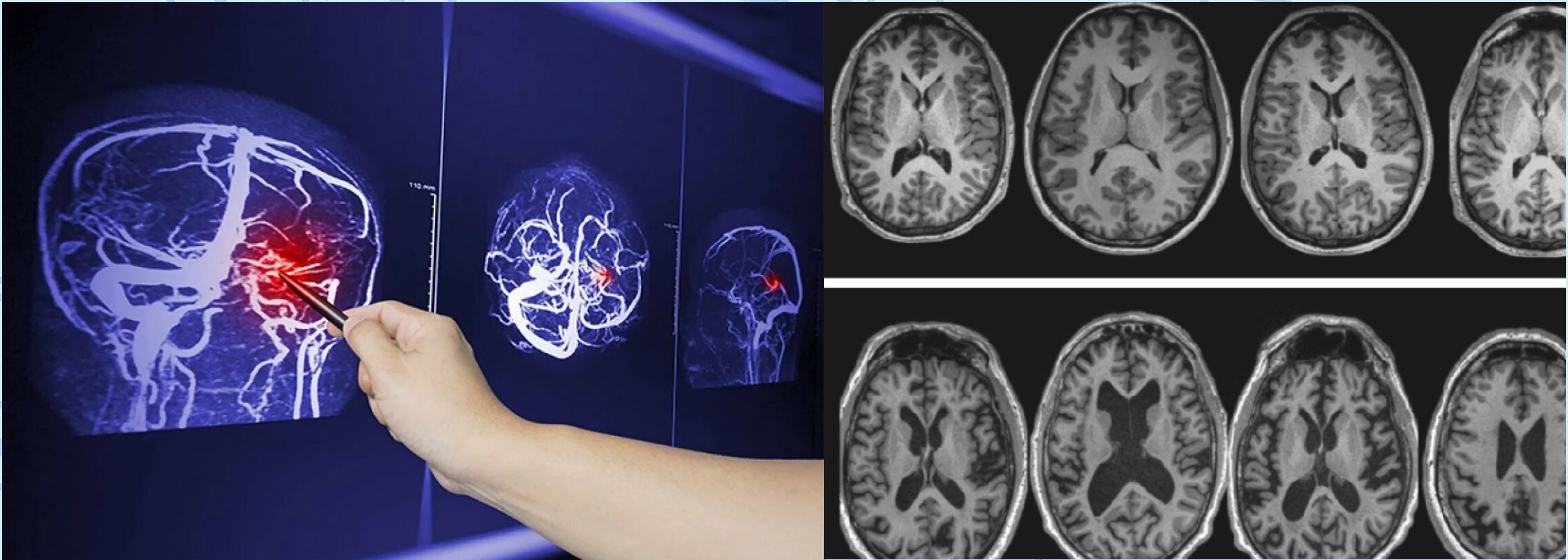
05

PART 07

Clinical Chill

Medical Cooling

Doctors plan to cool a brain-injured patient to 33°C for 24 hours. It's called "induced hypothermia". Why?



Cold Slows Damage

Metabolic Slowdown

Cooling the body to 33°C reduces metabolic rate, decreasing ATP consumption and slowing down harmful chemical reactions. This helps protect the brain from further damage. The deliberate temperature drop overrides the hypothalamus's normal temperature regulation, ensuring that the brain remains protected during the critical recovery period.





06

PART 05

Case Study

Stomach Emergency



A patient arrives at the hospital with vomiting and fever after eating meat left outside of a fridge for more than a day.

- What system in his body failed to maintain balance?
- Why did his body react by vomiting and heating up?
- How does this relate to homeostasis?

Emergency Homeostasis

After the digestive system failed to maintain balance, the hypothalamus triggered vomiting and fever to fight infection and flush out toxins. Raising core temperature helps inhibit microbes while vomiting expels contaminated food; both acute reactions illustrate homeostatic defence mechanisms that reestablish biochemical balance and protect systemic function.

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THANKS

$$E=mc^2$$